

Synthetic Data Augmentation via GANs for Rare Skin Disease Classification

(2023)

Problem

Addressing severe class imbalance for rare dermatological conditions.

Method

A conditional GAN generates synthetic dermoscopic images of rare conditions to augment training data.

Dataset

Augmented ISIC dataset plus a private rare-disease image set.

Evaluation

Downstream classifier accuracy improvement with vs without synthetic augmentation.

Results

Synthetic augmentation improved minority-class recall by 12 percentage points.

Limitations

Synthetic image realism was judged only qualitatively, not validated by dermatologists.